MIL-S-12613C
16 August 1982
SUPERSEDING
MIL-S-12613B(MU)
21 October 1969

MILITARY SPECIFICATION

SILICA, COLLOIDAL, SYNTHETIC

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of synthetic colloidal silica, hereinafter referred to as silica.

1.2 Classification. Silica shall be of the following types, as specified (see 6.2).

Type I - 0.1 percent maximum apparent density
Type II - 0.035 to 0.037 percent apparent density

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

B212 - Apparent Density of Free-Flowing Metal Powders
C169 - Chemical Analysis Of Soda-Lime And Borosilicate Glass
D1193 - Reagent Water
D1895 - Apparent Density, Bulk Factor, And Pourability Of Plastic Materials

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Silica shall conform to the chemical and physical characteristics of table I when tested as specified therein.

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TABLE I. Chemical and physical characteristics

Characteristic	Requirement		Test paragraph
	Type I	Type II	
Silicon dioxide, minimum percent by weight	95	95	4.2.4.1
Volatile matter, maximum percent by weight	5	5	4.2.4.2
Apparent density, grams per cubic centimeter	0.1 Max	0.035 to 0.037	4.2.4.3

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the silica of one type produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test (see 6.6). Sampling shall be conducted in accordance with table II. A representative specimen of approximately 1/2 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

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TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container damaged or leaking
(c) Marking incorrect, missing, or illegible
(d) Unitization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.6). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Silicon dioxide. Determine percent by weight of silicon dioxide in accordance with ASTM C169.

4.2.4.2 Volatile matter. Transfer approximately 5 grams of the specimen which has been weighed to the nearest milligram to a tared weighing bottle. Heat to constant weight at 150° ± 5°C, cool to room temperature in a desiccator, and weigh to the nearest milligram. Calculate the percent volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100\ (A - B)}{W}$$

where: A = Weight in grams of bottle and specimen before heating,
B = Weight in grams of bottle and specimen after heating, and
W = Weight in grams of specimen.

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4.2.4.3 Apparent density. Determine the apparent density of type I silica in accordance with ASTM D1895. Determine the apparent density of type II silica in accordance with ASTM B212.

5. PACKAGING

5.1 Unit packing, industrial. A quantity of silica, as specified (see 6.2), shall be unit packed industrially in accordance with MIL-STD-1188.

5.2 Packing, industrial. Silica, unit packed as specified in 5.1, shall be packed industrially in accordance with MIL-STD-1188.

5.3 Unitization. Packs of silica shall be unitized in a manner to assure safe delivery at destination, acceptance by common carrier, and conformance with Uniform Freight Classification Rules and National Motor Freight Classification Rules.

5.4 Marking. Containers shall be marked in accordance with MIL-STD-1188 and shall show date of manufacture and lot number of the silica.

6. NOTES

6.1 Intended use. Type I silica is intended for use as a dispersing and anticaking ingredient in the manufacture of riot control agents. Type II silica is intended for use as a thickening agent for epoxies and other liquid sealants and adhesives.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Type of silica required (see 1.2)
(c) Unit quantity required (see 5.2)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Silica types. Type I silica is the same material described in MIL-S-12613B(MU). Type II silica is a new type.

6.6 Sampling and testing precautions. Silica is a hazardous material if inhaled and contractors are advised to protect their workers with an approved dust face mask.

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MIL-S-12613C Silica, Colloidal, Synthetic

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